Section 4.4-Part 1

Orthonormal Bases

List of homework

Strang

Section 4.4: 2, 3, 6, 17, 15, 18, 21, 23, 24, 36

Bases for a vector space

Definitions:

- **Recall that a basis for a vector space is a linearly independent spanning set that spans the vector space.**
- **ANY linear independent spanning set is a basis.**
  - **For** $\mathbb{R}^2$, $\left\{\begin{bmatrix}15\\15\end{bmatrix}, \begin{bmatrix}15.1\\14.9\end{bmatrix}\right\}$, $\left\{\begin{bmatrix}10\\8\end{bmatrix}, \begin{bmatrix}0.02\\-0.01\end{bmatrix}\right\}$, **and** $\left\{\begin{bmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{bmatrix}, \begin{bmatrix}1/\sqrt{2}\\-1/\sqrt{2}\end{bmatrix}\right\}$ **are all bases.**
- **Many things become much much simpler our basis vectors are orthogonal, and even more so when they all have a length of "1".**

Orthogonal and orthonormal bases

Definitions:

- **An orthogonal basis is one where all the basis vectors are orthogonal to one another:**

$$\vec{v}_i^T \vec{v}_j = 0, \forall i \neq j$$

- **An orthonormal basis is one where all the basis vectors are orthogonal to one another AND they all have length (norm) of one:**

$$\vec{v}_i^T \vec{v}_j = 0, \forall i \neq j, \qquad \vec{v}_i^T \vec{v}_i = 1, \forall i$$

Examples of orthogonal matrices

Example 1 **(Rotation)** Q rotates every vector in the plane by the angle θ:

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \quad \text{and} \quad Q^{\mathrm{T}} = Q^{-1} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}.$$

Example 2 **(Permutation)** These matrices change the order to (y, z, x) and (y, x):

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y \\ z \\ x \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}.$$

Inverse = transpose:

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{z} \\ \boldsymbol{x} \end{bmatrix} = \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \\ \boldsymbol{z} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{x} \end{bmatrix} = \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{bmatrix}.$$

Every permutation matrix is an orthogonal matrix.

Example: Reflection matrix

Example 3 **(Reflection)** If $\boldsymbol{u}$ is any unit vector, set $Q = I - 2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$. Notice that $\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$ is a matrix while $\boldsymbol{u}^{\mathrm{T}}\boldsymbol{u}$ is the number $\|\boldsymbol{u}\|^2 = 1$. Then Q^{T} and Q^{-1} both equal Q:

$$Q^{\mathrm{T}} = I - 2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}} = Q \quad \text{and} \quad Q^{\mathrm{T}}Q = I - 4\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}} + 4\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}} = I. \tag{2}$$

Note: u is the direction vector for reflection.

If Q has orthonormal columns ($Q^{\mathrm{T}}Q = I$), it leaves lengths unchanged:

Same length for Qx $\quad \|Qx\| = \|x\|$ **for every vector** x. (3)

Q also preserves dot products: $(Qx)^{\mathrm{T}}(Qy) = x^{\mathrm{T}}Q^{\mathrm{T}}Qy = x^{\mathrm{T}}y$. Just use $Q^{\mathrm{T}}Q = I$!

Projections with an orthogonal basis.

- **Projections and pseudoinverses require a matrix inverse:**

$$proj_V(\vec{u}) = A\left(A^T A\right)^{-1} A^T \vec{u}, \qquad A^+ = \left(A^T A\right)^{-1} A^T$$

- **The columns of A are a basis for V:**

$$A = [\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n]$$

- **The entries of $A^T A$ are the dot products of the basis vectors:**

$$A^T A = \begin{bmatrix} \vec{v}_1^T \vec{v}_1 & \vec{v}_1^T \vec{v}_2 & \cdots & \vec{v}_1^T \vec{v}_n \\ \vec{v}_2^T \vec{v}_1 & \vec{v}_2^T \vec{v}_2 & \cdots & \vec{v}_2^T \vec{v}_n \\ \vdots & \vdots & & \vdots \\ \vec{v}_n^T \vec{v}_1 & \vec{v}_n^T \vec{v}_2 & \cdots & \vec{v}_n^T \vec{v}_n \end{bmatrix}$$

- **If the basis is orthogonal, $A^T A$ is a diagonal matrix; matrix inverse is easy!**

Projections with an orthogonal basis.

$$proj_V(\vec{u}) = A\left(A^T A\right)^{-1} A^T \vec{u}$$

$$A = [\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n], \qquad \vec{v}_i^T \vec{v}_j = 0, \forall i \neq j$$

$$A^T A = \begin{bmatrix} \vec{v}_1^T \vec{v}_1 & 0 & \cdots & 0 \\ 0 & \vec{v}_2^T \vec{v}_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \vec{v}_n^T \vec{v}_n \end{bmatrix} \rightarrow (A^T A)^{-1} = \begin{bmatrix} \frac{1}{\vec{v}_1^T \vec{v}_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\vec{v}_2^T \vec{v}_2} & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \frac{1}{\vec{v}_n^T \vec{v}_n} \end{bmatrix}$$

Projections with an orthogonal basis.

$$proj_V(\vec{u}) = A\left(A^T A\right)^{-1} A^T \vec{u}$$

$$(A^T A)^{-1} A^T \vec{u} = \begin{bmatrix} \frac{1}{\vec{v}_1^T \vec{v}_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\vec{v}_2^T \vec{v}_2} & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \frac{1}{\vec{v}_n^T \vec{v}_n} \end{bmatrix} \begin{bmatrix} \vec{v}_1^T \\ \vec{v}_2^T \\ \vdots \\ \vec{v}_n^T \end{bmatrix} \vec{u} = \begin{bmatrix} \frac{\vec{v}_1^T \vec{u}}{\vec{v}_1^T \vec{v}_1} \\ \frac{\vec{v}_2^T \vec{u}}{\vec{v}_2^T \vec{v}_2} \\ \vdots \\ \frac{\vec{v}_n^T \vec{u}}{\vec{v}_n^T \vec{v}_n} \end{bmatrix}$$

Projections with an orthogonal basis.

$$proj_V(\vec{u}) = A\left(A^T A\right)^{-1} A^T \vec{u}$$

$$A(A^T A)^{-1} A^T \vec{u} = [\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n] \begin{bmatrix} \frac{\vec{v}_1^T \vec{u}}{\vec{v}_1^T \vec{v}_1} \\ \frac{\vec{v}_2^T \vec{u}}{\vec{v}_2^T \vec{v}_2} \\ \vdots \\ \frac{\vec{v}_n^T \vec{u}}{\vec{v}_n^T \vec{v}_n} \end{bmatrix} = \frac{\vec{v}_1^T \vec{u}}{\vec{v}_1^T \vec{v}_1} \vec{v}_1 + \frac{\vec{v}_2^T \vec{u}}{\vec{v}_2^T \vec{v}_2} \vec{v}_2 + \frac{\vec{v}_n^T \vec{u}}{\vec{v}_n^T \vec{v}_n} \vec{v}_n$$

- **When we have an orthogonal basis, the projection onto the vector space is simply the sum of the projections onto each basis vector!**

Projections with an orthonormal basis.

- **For an orthonormal basis:**

$$\vec{v}_i^T \vec{v}_j = 0, \forall i \neq j, \qquad \vec{v}_i^T \vec{v}_i = 1, \forall i$$

- $A^T A$ **is an identity matrix!:**

$$A^T A = \begin{bmatrix} \vec{v}_1^T \vec{v}_1 & \vec{v}_1^T \vec{v}_2 & \cdots & \vec{v}_1^T \vec{v}_n \\ \vec{v}_2^T \vec{v}_1 & \vec{v}_2^T \vec{v}_2 & \cdots & \vec{v}_2^T \vec{v}_n \\ \vdots & \vdots & & \vdots \\ \vec{v}_n^T \vec{v}_1 & \vec{v}_n^T \vec{v}_2 & \cdots & \vec{v}_n^T \vec{v}_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = I$$

- **Projections and pseudoinverses are greatly simplified:**

$$proj_V(\vec{u}) = A\left(A^T A\right)^{-1} A^T \vec{u} \rightarrow proj_V(\vec{u}) = AA^T \vec{u}$$

$$A^+ = \left(A^T A\right)^{-1} A^T \rightarrow A^+ = A^T$$

Projections with an orthonormal basis.

$$proj_V(\vec{u}) = A\left(A^T A\right)^{-1} A^T \vec{u} = AA^T \vec{u}$$

$$AA^T \vec{u} = [\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n] \begin{bmatrix} \vec{v}_1^T \vec{u} \\ \vec{v}_2^T \vec{u} \\ \vdots \\ \vec{v}_n^T \vec{u} \end{bmatrix} = (\vec{v}_1^T \vec{u})\vec{v}_1 + (\vec{v}_2^T \vec{u})\vec{v}_2 + (\vec{v}_n^T \vec{u})\vec{v}_n$$

- **When we have an orthonormal basis, the projection onto the vector space is simply the sum of the projections onto each basis vector, and each projection is just the vector scaled by a single dot product.**

Section 4.4-Part 2

Gram-Schmidt Orthonormalization

How to find an orthonormal basis?

The Gram-Schmidt algorithm: **A robust method to generate an orthonormal basis starting from any basis.**

Key ideas:

- **The projection can be used express a vector into a parallel and orthogonal component, relative to some vector space.**
- **Projections onto vector space are easy to evaluate when we are working with an orthogonal basis set; they are sums of projections onto of individual vectors.**
- **Dividing by the length of a vector yields a unit vector in the same direction.**

Gram-Schmidt

Goal: Starting with any basis for a vector space, $\{\vec{v_i}\}$ and create a new basis $\{\vec{w}_i\}$ that is orthogonal!

Method:

- **The first vector of the new basis is just the first vector of the original basis; this vector spans a line in the original vector space:**

$$\vec{w}_1 = \vec{v}_1, \qquad W_1 = span(\{\vec{w}_1\})$$

- **We construct a vector that is orthogonal to this line by finding the orthogonal component of the second vector from the original set; we now have an orthogonal pair of vector that span a plane in the original vector space:**

$$\vec{w}_2 = \vec{v}_2 - proj_{W_1}(\vec{v}_2) = \vec{v}_2 - \frac{\vec{w}_1^T \vec{v}_2}{\vec{w}_1^T \vec{w}_1}\vec{w}_1, \qquad W_2 = span(\{\vec{w}_1, \vec{w}_2\})$$

Gram-Schmidt

Method, continued:

- **We construct a vector that is orthogonal to this plane by finding the orthogonal component of the third vector from the original set; we now have an orthogonal triplet of vector that spans a 3D subspace of the original vector space:**

$$\vec{w}_3 = \vec{v}_3 - proj_{W_2}(\vec{v}_3) = \vec{v}_3 - \frac{\vec{w}_1^T \vec{v}_3}{\vec{w}_1^T \vec{w}_1}\vec{w}_1 - \frac{\vec{w}_2^T \vec{v}_3}{\vec{w}_2^T \vec{w}_2}\vec{w}_2, \qquad W_3 = span(\{\vec{w}_1, \vec{w}_2, \vec{w}_3\})$$

- **We continue as needed, until we reach the dimension of the space. At each step we add another orthogonal component:**

$$\vec{w}_i = \vec{v}_i - proj_{W_{i-1}}(\vec{v}_i) = \vec{v}_i - \frac{\vec{w}_1^T \vec{v}_i}{\vec{w}_1^T \vec{w}_1}\vec{w}_1 - \frac{\vec{w}_2^T \vec{v}_i}{\vec{w}_2^T \vec{w}_2}\vec{w}_2 - \cdots - \frac{\vec{w}_{i-1}^T \vec{v}_i}{\vec{w}_{i-1}^T \vec{w}_{i-1}}\vec{w}_{i-1}$$

Gram-Schmidt

Normalization

- **We now have an orthogonal basis:**

$$V = span(\{\vec{w}_1, \vec{w}_2, \cdots, \vec{w}_n\}), \qquad \vec{w}_i^T \vec{w}_j = 0, \forall i \neq j$$

- **We obtain an orthonormal basis by dividing each basis vector by its length:**

$$\vec{q}_i = \left(\frac{1}{\|\vec{w}_i\|}\right)\vec{w}_i, \forall i$$

$$V = span(\{\vec{q}_1, \vec{q}_2, \cdots, \vec{q}_n\}), \qquad \vec{q}_i^T \vec{q}_j = 0, \forall i \neq j, \qquad \|\vec{q}_i\| = 1, \forall i$$

Gram-Schmidt

An example: $S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 4 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 4 \\ 3 \\ 0 \end{pmatrix} \right\}, V = span(S)$

$$\vec{w}_1 = \vec{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \qquad \vec{w}_2 = \vec{v}_2 - \frac{\vec{w}_1^T \vec{v}_2}{\vec{w}_1^T \vec{w}_1} \vec{w}_1 = \begin{pmatrix} 1 \\ 2 \\ 4 \\ 2 \\ 1 \end{pmatrix} - \left(\frac{10}{5}\right) \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 2 \\ 0 \\ -1 \end{pmatrix}$$

Gram-Schmidt

$$S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 4 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 4 \\ 3 \\ 0 \end{pmatrix} \right\}, V = span(S) \qquad W_2 = span\left(\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 2 \\ 0 \\ -1 \end{pmatrix} \right\} \right)$$

$$\vec{w}_3 = \vec{v}_3 - \frac{\vec{w}_1^T \vec{v}_3}{\vec{w}_1^T \vec{w}_1} \vec{w}_1 - \frac{\vec{w}_2^T \vec{v}_3}{\vec{w}_2^T \vec{w}_2} \vec{w}_2 = \begin{pmatrix} 0 \\ 3 \\ 4 \\ 3 \\ 0 \end{pmatrix} - \left(\frac{10}{5}\right) \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} - \left(\frac{8}{6}\right) \begin{pmatrix} -1 \\ 0 \\ 2 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -2/3 \\ 1 \\ -2/3 \\ 1 \\ -2/3 \end{pmatrix}$$

Gram-Schmidt

$$S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 4 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 4 \\ 3 \\ 0 \end{pmatrix} \right\}, V = span(S) \qquad W_2 = span\left(\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 2 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -2/3 \\ 1 \\ -2/3 \\ 1 \\ -2/3 \end{pmatrix} \right\} \right)$$

$$\vec{q}_i = \left(\frac{1}{\|\vec{w}_i\|}\right)\vec{w}_i \rightarrow \vec{q}_1 = \left(\frac{1}{\sqrt{5}}\right)\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \vec{q}_2 = \left(\frac{1}{\sqrt{6}}\right)\begin{pmatrix} -1 \\ 0 \\ 2 \\ 0 \\ -1 \end{pmatrix}, \vec{q}_3 = \left(\frac{1}{\sqrt{30}}\right)\begin{pmatrix} -2 \\ 3 \\ -2 \\ 3 \\ -2 \end{pmatrix}$$

Examples

Orthonormal bases

Ex 1: Projections with orthonormal bases.

- **Find the projection of the vector $\vec{u}$ onto the subspace of spanned by $\{\vec{q}_1, \vec{q}_2, \vec{q}_3, \vec{q}_4\}$ and, where:**

$$\vec{u} = (1,2,3,4,5)$$

$$\vec{q}_1 = \left(\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}\right), \vec{q}_2 = (0,0,1,0,0),$$

$$\vec{q}_3 = \left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0,0,0\right), \vec{q}_4 = \left(0,0,0, \frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$$

Projections with orthonormal bases.

$$Q = [\vec{q}_1 \quad \vec{q}_2 \quad \vec{q}_3 \quad \vec{q}_4]$$

$$proj_V(\vec{u}) = Q\left(Q^T Q\right)^{-1} Q^T \vec{u}$$

$$Q = \begin{bmatrix} 1/2 & 0 & 1/\sqrt{2} & 0 \\ 1/2 & 0 & -1/\sqrt{2} & 0 \\ 0 & 1 & 0 & 0 \\ 1/2 & 0 & 0 & 1/\sqrt{2} \\ 1/2 & 0 & 0 & -1/\sqrt{2} \end{bmatrix} \rightarrow Q^T Q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow proj_V(\vec{u}) = Q Q^T \vec{u}$$

Projections with orthonormal bases.

$$proj_V(\vec{u}) = QQ^T\vec{u}$$

$$Q^T\vec{u} = \begin{bmatrix} 1/2 & 1/2 & 0 & 1/2 & 1/2 \\ 0 & 0 & 1 & 0 & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ -1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

Projections with orthonormal bases.

$$proj_V(\vec{u}) = QQ^T\vec{u} = \begin{bmatrix} 1/2 & 0 & 1/\sqrt{2} & 0 \\ 1/2 & 0 & -1/\sqrt{2} & 0 \\ 0 & 1 & 0 & 0 \\ 1/2 & 0 & 0 & 1/\sqrt{2} \\ 1/2 & 0 & 0 & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 6 \\ 3 \\ -1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 5/2 \\ 7/2 \\ 3 \\ 5/2 \\ 7/2 \end{bmatrix}$$

Ex 2: Gram-Schmidt orthonormalization.

- **Find an orthonormal basis for the vector space:**

$$V = span(\{\vec{v}_1, \vec{v}_2, \vec{v}_3\})$$

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \vec{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}$$

Gram-Schmidt orthonormalization.

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \vec{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}$$

- **First generate an orthogonal basis:**

$$\vec{w}_1 = \vec{v}_1$$

$$\vec{w}_2 = \vec{v}_2 - \frac{\vec{w}_1^T \vec{v}_2}{\vec{w}_1^T \vec{w}_1} \vec{w}_1$$

$$\vec{w}_3 = \vec{v}_3 - \frac{\vec{w}_1^T \vec{v}_3}{\vec{w}_1^T \vec{w}_1} \vec{w}_1 - \frac{\vec{w}_2^T \vec{v}_3}{\vec{w}_2^T \vec{w}_2} \vec{w}_2$$

Gram-Schmidt orthonormalization.

$$\vec{v}_1 = \begin{bmatrix}1\\1\\1\\1\end{bmatrix}, \vec{v}_2 = \begin{bmatrix}2\\2\\1\\1\end{bmatrix}, \vec{v}_3 = \begin{bmatrix}1\\2\\2\\1\end{bmatrix}$$

- **First generate an orthogonal basis:**

$$\vec{w}_1 = \vec{v}_1 = \begin{bmatrix}1\\1\\1\\1\end{bmatrix}, \vec{w}_1^T\vec{w}_1 = 4, \vec{w}_1^T\vec{v}_2 = 6,$$

$$\vec{w}_2 = \vec{v}_2 - \frac{\vec{w}_1^T\vec{v}_2}{\vec{w}_1^T\vec{w}_1}\vec{w}_1 = \begin{bmatrix}2\\2\\1\\1\end{bmatrix} - \left(\frac{6}{4}\right)\begin{bmatrix}1\\1\\1\\1\end{bmatrix} = \begin{bmatrix}1/2\\1/2\\-1/2\\-1/2\end{bmatrix} \rightarrow \begin{bmatrix}1\\1\\-1\\-1\end{bmatrix}$$

Gram-Schmidt orthonormalization.

$$\vec{v}_1 = \begin{bmatrix}1\\1\\1\\1\end{bmatrix}, \vec{v}_2 = \begin{bmatrix}2\\2\\1\\1\end{bmatrix}, \vec{v}_3 = \begin{bmatrix}1\\2\\2\\1\end{bmatrix}$$

- **First generate an orthogonal basis:**

$$\vec{w}_1 = \begin{bmatrix}1\\1\\1\\1\end{bmatrix}, \vec{w}_2 = \begin{bmatrix}1\\1\\-1\\-1\end{bmatrix}, \vec{w}_1^T\vec{w}_1 = 4, \vec{w}_1^T\vec{v}_3 = 6, \vec{w}_2^T\vec{w}_2 = 4, \vec{w}_2^T\vec{v}_3 = 0$$

$$\vec{w}_3 = \vec{v}_3 - \frac{\vec{w}_1^T\vec{v}_3}{\vec{w}_1^T\vec{w}_1}\vec{w}_1 - \frac{\vec{w}_2^T\vec{v}_3}{\vec{w}_2^T\vec{w}_2}\vec{w}_2 = \begin{bmatrix}1\\2\\2\\1\end{bmatrix} - \left(\frac{6}{4}\right)\begin{bmatrix}1\\1\\1\\1\end{bmatrix} - \left(\frac{0}{4}\right)\begin{bmatrix}1\\1\\-1\\-1\end{bmatrix} = \begin{bmatrix}-1/2\\1/2\\1/2\\-1/2\end{bmatrix} \rightarrow \begin{bmatrix}-1\\1\\1\\-1\end{bmatrix}$$

Gram-Schmidt orthonormalization.

$$\vec{w}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \vec{w}_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \vec{w}_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

- **Then normalize:**

$$\vec{q}_i = \left(\frac{1}{\|\vec{w}_i\|}\right)\vec{w}_i, \|\vec{w}_i\| = \sqrt{\vec{w}_i^T \vec{w}_i}$$

$$\vec{w}_1^T \vec{w}_1 = 4, \qquad \vec{w}_2^T \vec{w}_2 = 4, \qquad \vec{w}_3^T \vec{w}_3 = 4$$

$$\vec{q}_1 = \left(\frac{1}{2}\right)\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \vec{q}_2 = \left(\frac{1}{2}\right)\begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \\ -1/2 \\ -1/2 \end{bmatrix}, \vec{q}_3 = \left(\frac{1}{2}\right)\begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/2 \\ 1/2 \\ -1/2 \end{bmatrix}$$

Gram-Schmidt orthonormalization.

- **Our orthonormal basis is:**

$$Q = \{\vec{q}_1, \vec{q}_2, \vec{q}_3\}, \qquad V = span(Q)$$

$$\vec{q}_1 = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \qquad \vec{q}_2 = \begin{bmatrix} 1/2 \\ 1/2 \\ -1/2 \\ -1/2 \end{bmatrix}, \qquad \vec{q}_3 = \begin{bmatrix} -1/2 \\ 1/2 \\ 1/2 \\ -1/2 \end{bmatrix}$$

Section 4.4-Part 3

The QR decomposition

QR Decomposition

- **If we organize our basis vectors as the columns of a matrix, we can summarize Gram-Schmidt as:**

$$A \rightarrow W \rightarrow Q$$

$$[\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n] \rightarrow [\vec{w}_1 \quad \vec{w}_2 \quad \cdots \quad \vec{w}_n] \rightarrow [\vec{q}_1 \quad \vec{q}_2 \quad \cdots \quad \vec{q}_n]$$

- **All share the same column space:**

$$V = \boldsymbol{C}(A) = \boldsymbol{C}(W) = \boldsymbol{C}(Q)$$

- **The Gram-Schmidt process involves taking linear combinations of columns of A**

$$\vec{w}_i = \vec{v}_i - proj_{W_{i-1}}(\vec{v}_i) = \vec{v}_i - \frac{\vec{w}_1^T \vec{v}_i}{\vec{w}_1^T \vec{w}_1}\vec{w}_1 - \frac{\vec{w}_2^T \vec{v}_i}{\vec{w}_2^T \vec{w}_2}\vec{w}_2 - \cdots - \frac{\vec{w}_{i-1}^T \vec{v}_i}{\vec{w}_{i-1}^T \vec{w}_{i-1}}\vec{w}_{i-1}$$

$$\vec{q}_i = \left(\frac{1}{\|\vec{w}_i\|}\right)\vec{w}_i, \forall i$$

QR Decomposition

$$A \to W \to Q$$

$$[\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n] \to [\vec{w}_1 \quad \vec{w}_2 \quad \cdots \quad \vec{w}_n] \to [\vec{q}_1 \quad \vec{q}_2 \quad \cdots \quad \vec{q}_n]$$

$$\vec{w}_i = \vec{v}_i - proj_{W_{i-1}}(\vec{v}_i) = \vec{v}_i - \frac{\vec{w}_1^T \vec{v}_i}{\vec{w}_1^T \vec{w}_1}\vec{w}_1 - \frac{\vec{w}_2^T \vec{v}_i}{\vec{w}_2^T \vec{w}_2}\vec{w}_2 - \cdots - \frac{\vec{w}_{i-1}^T \vec{v}_i}{\vec{w}_{i-1}^T \vec{w}_{i-1}}\vec{w}_{i-1}$$

$$\vec{q}_i = \left(\frac{1}{\|\vec{w}_i\|}\right)\vec{w}_i, \forall i$$

Elementary column operations:

- **Addition of a scalar multiple of one column to another (repeated).**
- **Multiplication of a column by a scalar.**

QR Decomposition

Just as elementary row operations can be represented by matrix multiplication, by a nonsingular matrix on the left.

$$A' = EA$$

Elementary column operations can be represented by matrix multiplication, by a nonsingular elementary matrix on the right:

$$A' = AE$$

QR Decomposition

$$A \to W \to Q$$

$$[\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n] \to [\vec{w}_1 \quad \vec{w}_2 \quad \cdots \quad \vec{w}_n] \to [\vec{q}_1 \quad \vec{q}_2 \quad \cdots \quad \vec{q}_n]$$

- **W is obtained from A by the repeated addition of a scalar multiple of one column to another.**

$$A \to W, \qquad W = AX$$

- **Multiplication of a column by a scalar.**

$$W \to Q, \qquad Q = WY$$

QR Decomposition

Elementary column operations can be represented by matrix multiplication.

- **On the right, by a nonsingular elementary matrix:** $A' = AE$

$$A \rightarrow W \rightarrow Q$$

- W **is obtained from** A **by the repeated addition of a scalar multiple of one column to another.**

$$W = AX$$

- Q **is obtained from** W **by multiplication of each column by a scalar.**

$$Q = WY$$

$$Q = AXY \rightarrow A = Q(XY)^{-1} = QR$$

QR Decomposition

$$A \to W \to Q$$

$$W = AX, Q = WY \to A = Q(XY)^{-1} = QR$$

- Q **has orthonormal columns:**

$$Q^T Q = I$$

$$A = QR \to R = Q^T A$$

- R **is easy to compute!**
  - $r_{ij} = \vec{q}_i^T \vec{v}_j$, **but** $\vec{q}_i$ **is constructed to be orthogonal to all preceeding** $\vec{v}_j$
  - $\vec{q}_i^T \vec{a}_j = 0, \forall i > j$ → R **is an upper triangular matrix.**

Least squares $R^{\mathrm{T}}R\widehat{\boldsymbol{x}} = R^{\mathrm{T}}Q^{\mathrm{T}}\boldsymbol{b}$ or $R\widehat{\boldsymbol{x}} = Q^{\mathrm{T}}\boldsymbol{b}$ or $\boxed{\widehat{\boldsymbol{x}} = R^{-1}Q^{\mathrm{T}}\boldsymbol{b}}$ (10)

Any m by n matrix A with independent columns can be factored into $A = QR$. The m by n matrix Q has orthonormal columns, and the square matrix R is upper triangular with positive diagonal. We must not forget why this is useful for least squares: $\boldsymbol{A^{\mathrm{T}}A} = \boldsymbol{(QR)^{\mathrm{T}}QR} = \boldsymbol{R^{\mathrm{T}}Q^{\mathrm{T}}QR} = \boldsymbol{R^{\mathrm{T}}R}$. The least squares equation $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$ simplifies to $R^{\mathrm{T}}R\widehat{\boldsymbol{x}} = R^{\mathrm{T}}Q^{\mathrm{T}}\boldsymbol{b}$. Then finally we reach $R\widehat{\boldsymbol{x}} = Q^{\mathrm{T}}\boldsymbol{b}$: good.

- **For any full column rank matrix we can define a pseudoinverse:**

$$\mathrm{rank}(A) = n, A \in \mathbb{R}^{m\times n} \rightarrow A^{+} = \left(A^{T}A\right)^{-1}A^{T}$$

- **The pseudoinverse gives us the "best" answer for any right-hand side:**

$$A\vec{x} = \vec{b} \rightarrow \vec{x} = A^{+}\vec{b}, \forall\vec{b}$$

$$A = QR \rightarrow A^{T}A = (QR)^{T}QR = R^{T}Q^{T}QR = R^{T}R$$

$$A^{+} = \left(A^{T}A\right)^{-1}A^{T} = \left(R^{T}R\right)^{-1}R^{T}Q^{T} = R^{-1}\left(R^{T}\right)^{-1}R^{T}Q^{T} = R^{-1}Q^{T}$$

$$A^{+} = R^{-1}Q^{T}$$

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & -3 \\ 0 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ -1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ 0 & -2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{\mathbf{2}} & \sqrt{2} & \sqrt{18} \\ \mathbf{0} & \sqrt{\mathbf{6}} & -\sqrt{6} \\ \mathbf{0} & \mathbf{0} & \sqrt{\mathbf{3}} \end{bmatrix} = QR.$$

Formula: $A = [\overrightarrow{v_1} \quad \overrightarrow{v_2} \quad \overrightarrow{v_3}] = [\overrightarrow{q_1} \quad \overrightarrow{q_2} \quad \overrightarrow{q_3}] \begin{bmatrix} \overrightarrow{q_1^T}\overrightarrow{v_1} & \overrightarrow{q_1^T}\overrightarrow{v_2} & \overrightarrow{q_1^T}\overrightarrow{v_3} \\ 0 & \overrightarrow{q_2^T}\overrightarrow{v_2} & \overrightarrow{q_2^T}\overrightarrow{v_3} \\ 0 & 0 & \overrightarrow{q_3^T}\overrightarrow{v_3} \end{bmatrix}$

Why?

Because A= QR from Gram-Schmidt, it follows know that R= $Q^T A$ by multiplying Q^T.